Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Advanced Manufacturing Technologies for Next-Generation Batteries*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility (), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (), Cybersecurity (), AI (), Next-gen Communication (), Advanced Robotics & Manufacturing (), Quantum ()			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	*Materials Processing Technologies*	*Metal Materials*	*Energy Materials Technologies*
Concept of Proposed Technology	Battery manufacturing plays a pivotal role in enabling the global energy transition, strengthening strategic autonomy, and supporting industrial competitiveness across mobility, aerospace, defence, and renewable energy sectors. Today's rapidly rising performance and market requirements—higher energy density, reduced cost, faster charging, and increasingly strict environmental regulations— are pushing conventional battery chemistries and manufacturing processes to their limits. Next generation anode chemistries such as silicon and lithium metal offer step changes in gravimetric and volumetric energy density and decrease reliance on graphite, a supply chain dominated by China. However, each faces major barriers to commercialization: silicon suffers from extreme volume expansion during cycling, while lithium metal introduces challenges in handling, integration, dendrite suppression, and surface stability.			

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	Simultaneously, today's wet electrode manufacturing for battery cathodes remains one of the most energy, space, and cost intensive steps in cell production. The dependence on toxic NMP, large drying ovens, and full dry room environments leads to high CAPEX and OPEX, while slow drying kinetics limit electrode loading and cap achievable energy density. Dry electrode coating, a solvent free and mechanically driven process, has emerged as a transformative solution capable of enabling thicker electrodes, reducing costs, minimizing environmental footprint, and enhancing industrial scalability. Yet robust, high throughput industrial dry coating technologies—especially PFAS free—are not widely available. As next generation chemistries mature (Si, Li metal, anode free, and solid-state architectures), the role of the copper current collector is also being redefined. High performance systems increasingly demand lightweight, structured, and functionalized current collectors that improve ion transport, reduce resistance, and enhance mechanical integrity. These advanced architectures must be compatible with automated, high throughput R2R manufacturing, a core competency at Holst Centre. Holst Centre-TNO is a leading Dutch R&D institute specializing in printed electronics, flexible device integration, and advanced battery manufacturing technologies with comprehensive research and development infrastructure. This proposal introduces an integrated suite of materials, processes, and equipment innovations—spanning dry coating, solid state battery integration, Li metal stabilization, and structured current collectors—to unlock practical, scalable commercialization of next generation battery technologies with Korean SMEs.
Rationale for Joint R&D Project	Korea is a global leader in battery materials, cell manufacturing, and large-scale production infrastructure. However, the global battery industry is now undergoing a structural transition toward next-generation battery architectures and manufacturing methods. To remain competitive—especially against rapidly scaling LFP platforms—future battery systems must deliver higher performance at significantly lower cost. Achieving this requires entirely new manufacturing paradigms, advanced interfacial control methods, and novel equipment platforms that extend far beyond incremental improvements to existing lithium-ion processes. At the same time, intensifying global competition and the dominance of Chinese battery supply chains present shared strategic challenges for both Korea and the European Union. These include supply-chain resilience, access to next-generation technologies, and the ability to industrialize advanced battery systems beyond current incumbent platforms. A joint R&D program between Korean SMEs and Holst Centre directly addresses these challenges by combining Korea's manufacturing strengths and industrial agility with the Netherlands' capabilities in advanced materials development, equipment innovation, and sustainable manufacturing processes. Although materials innovation is progressing rapidly at the laboratory scale, the industrialization of next-generation batteries remains a major bottleneck. Successful commercialization requires deep expertise in thin- and thick-film processing, interface and barrier engineering, high-throughput materials structuring, roll-to-roll integration, and precision manufacturing—competencies that typically exceed the scope of any single national ecosystem. This gap highlights the need for international collaboration where complementary capabilities can be combined to accelerate scalability. Holst Centre brings decades of applied R&D experience in thin- and thick-film integration, roll-to-roll

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	manufacturing, dry-coating technologies, and advanced barrier and electrode engineering. These capabilities are supported by extensive pre-pilot infrastructure, advanced metrology, environmental and reliability testing, and full electrochemical characterization facilities. Holst Centre also offers more than a decade of solid-state battery development experience, including successful technology maturation programs and multiple spin-out companies, providing proven insights into scale-up pathways, rapid prototyping, and time-to-market acceleration. Positioned at the core of the Dutch and broader European battery ecosystem, Holst Centre can connect Korean SMEs with European suppliers, equipment makers, integrators, and potential customers. This combination of ecosystem access and joint technical development enhances the global competitiveness and commercial readiness of participating SMEs. Through this joint R&D program, Korean SMEs and Holst Centre will co-develop scalable, high-performance, and sustainable battery manufacturing technologies that neither party could achieve independently reducing industrialization risk, accelerating commercialization timelines, and strengthening Korea's leadership in the next stage of global energy-storage innovation.
Key Words	1. PFAS-free dry-coating for battery electrode manufacturing 2. Silicon & lithium-metal anodes 3. Solid-state battery integration 4. Structured current collectors. 5. Thin/thick films integration

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	TRL3	End of the Project	6 TRL6
R&D Objectives	1. **Develop PFAS-free dry---coating electrode technologies:** • Formulate binder systems, optimize powder morphology, and engineer robust dry mixing & compaction processes for Li--ion, Na--ion, and solid---state systems. • Demonstrate stable, high-loading cathodes/anodes using solvent---free methods. **2. Advance solid--state battery materials and integration processes:** • Develop solid electrolytes, interfacial layers, and scalable integration routes compatible with R2R processing. • Validate interface engineering methods for improved interface contact and suppression of interfacial degradation. **3. Develop lithium -metal anode stabilization technologies:** • Engineer barrier layers, protective coatings, and handling strategies for Li-metal -roll-to-roll- integration. • Demonstrate improved cycling stability and manufacturability. **4. Develop advanced structured current collectors:** • Design lightweight, micro-structured copper and hybrid composite current collectors for Si and Li-metal anodes. • Establish scalable fabrication processes and equipment concepts.			
R&D Target Specifications	The target performance specifications for this joint R&D program will be defined at a high level to support the overall objectives of enabling scalable, high-performance dry-coating, solid-state battery integration, lithium-metal stabilization, and advanced structured current-collector technologies. Because the performance requirements vary significantly by application (e-mobility, aviation, stationary storage, high-power applications, etc.), detailed KPIs, quantitative targets, and measurable parameters will be co-developed with the participating Korean SMEs. These KPIs will be finalized during the early phase of the project based on: • The intended use-case requirements, • Materials selection, • Manufacturing constraints and commercial performance targets of each SME partner. <u>The high-level goals that will guide KPI definition include:</u> • Demonstrating scalable PFAS-free dry-coating with competitive mechanical, electrochemical, and processing performance relative to wet-coated electrodes. • Achieving improved interfacial stability and manufacturable integration routes for solid-state battery components. • Establishing efficient, stable, and production-ready Li-metal anode technologies suitable for R2R processing. • Demonstrating lightweight, structured current collectors that enable higher energy density and improved cycling performance.			

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	Final quantitative specifications (e.g., areal capacity, ionic conductivity, interfacial resistance, cycle retention, mechanical adhesion, throughput, etc.) will be jointly set with the SMEs to match their targeted product lines and market segments.
R&D Plan	The detailed R&D plan including specific tasks, methodologies, equipment configurations, and validation protocols will be co-developed with the participating Korean SMEs.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Materials, Process, and Equipment Development: Develop and optimize materials formulations, scalable process flows, and conceptual equipment designs relevant to dry-coating, solid-state assembly, Li-metal handling, and structured current collectors. • Benchmarking & State-of-the-Art Analysis: Conduct literature surveys, technology landscaping, and benchmarking against global state-of-the-art manufacturing approaches to inform strategic development directions. • Advanced Characterization & Post-Mortem Analysis: Conduct physical, chemical, morphological, and structural characterization (e.g.: SEM, XRF, XRD, XPS, AFM, mechanical testing, etc.). • Electrochemical Performance Evaluation: Build and test coin cells and pouch-cell prototypes, evaluate rate capability, cycling stability, interface resistance, and other performance metrics. • Pre-Pilot Infrastructure Access: Provide access to R2R thin/thick-film equipment, dry-coating lines, environmental chambers, metrology platforms, and reliability testing facilities. Also, can host the SME team as industrial residents. • Business & Market Intelligence: Support SMEs with high-level market analysis, competitive benchmarking, and technology-to-market strategies aligned with EU and global industry trends. • Dissemination & Exploitation Strategy: Lead knowledge dissemination, technical reporting, stakeholder engagement, and strategy for exploitation of results, including pathways to commercialization. • Ecosystem Facilitation: Connect SMEs with European suppliers, equipment makers, integrators, and potential customers. • Project Management & Technical Coordination: Lead project planning, milestone tracking, risk management, and overall coordination of tasks across consortium partners.
Expected R&R of Korean SME Partner	• Provide technology contributions as an equipment supplier (complete systems or components), materials supplier, component manufacturer, or co-development partner. • Deliver SME-specific requirements including target KPIs, performance metrics, integration constraints, and application-driven use-case needs. • Support development tasks such as materials input, prototype fabrication, and process adaptation for their manufacturing environment. • Conduct testing and validation of jointly developed materials, components, or processes within their internal

	facilities or pilot lines. • Contribute engineering insights on manufacturability, scalability, and cost targets relevant to Korean industrial deployment. • Ensure resource commitments (materials, engineering support, equipment access) necessary for iterative development and evaluation. • Engage in active collaboration with Holst Centre for refinement, troubleshooting, and optimization of integrated technologies. • SMEs will support joint dissemination and commercialization planning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Validated PFAS-free dry-coating technology for high-loading electrodes. • Scalable processes for solid-state battery integration with improved interfaces. • Next-generation lithium-metal stabilization and handling technologies. • Lightweight, structured current collectors enabling higher energy density. **Commercial & Industrial Impact** • Significant reduction in CAPEX/OPEX for electrode manufacturing (through solvent elimination, reduced drying, reduced dry-room footprint). • Accelerated technology transfer enabling SMEs to enter next-gen battery markets early. • Strengthened Korean–EU collaboration and supply-chain diversification. • Enhanced competitiveness against Chinese-dominated supply chains. • Ability for Korean SMEs to position themselves as global leaders in high-energy, sustainable, and cost-competitive battery manufacturing. **Broader Impact** • Supports (inter)national strategies for green manufacturing and advanced materials. • Creates pathways for future joint commercialization, licensing, and spin-off opportunities. • Contributes to global energy-transition goals through safer, cleaner, and higher-performance batteries.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems. 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization